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Stop Probing, Start Coding: Why Linear Probes and Sparse Autoencoders Fail at Compositional Generalisation

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Abstract

The linear representation hypothesis states that neural network activations encode high-level concepts as linear mixtures. However, under superposition, this encoding is a projection from a higher-dimensional concept space into a lower-dimensional activation space, and a linear decision boundary in the concept space need not remain linear after projection. In this setting, classical sparse coding methods with per-sample iterative inference leverage compressed sensing guarantees to recover latent factors. Sparse autoencoders (SAEs), on the other hand, amortise sparse inference into a fixed encoder, introducing a systematic gap. We show this amortisation gap persists across training set sizes, latent dimensions, and sparsity levels, causing SAEs to fail under out-of-distribution (OOD) compositional shifts. Through controlled experiments that decompose the failure, we identify *dictionary learning*—not the inference procedure—as the binding constraint: SAE-learned dictionaries point in substantially wrong directions, and replacing the encoder with per-sample FISTA on the same dictionary does not close the gap. An oracle baseline proves the problem is solvable with a good dictionary at all scales tested. Our results reframe the SAE failure as a dictionary learning challenge, not an amortisation problem, and point to scalable dictionary learning as the key open problem for sparse inference under superposition.

1 INTRODUCTION

Understanding the internal representations of Large Language Models (LLMs) is crucial for their safe and reliable deployment. The *linear representation hypothesis* (LRH) is a foundational assumption in mechanistic interpretability, stating that a model's activations are linear mixtures of underlying concepts (Jiang et al., 2024; Park et al., 2024; Smith, 2024). It has motivated crucial progress on methods such as linear probing for concept discovery and activation steering (Turner et al., 2023; Chalnev et al., 2024).

To make this precise, let $\mathbf{z} \in \mathbb{R}^{d_z}$ denote ground-truth latent variables (concepts), and let $\mathbf{y} \in \mathbb{R}^{d_y}$ denote a model's activations. The LRH asserts that the relationship between concepts and activations is linear: $\mathbf{y} \approx \mathbf{W}\mathbf{z}$ for some matrix $\mathbf{W} \in \mathbb{R}^{d_y \times d_z}$. The goal of interpretability is to recover $\mathbf{z}$ from $\mathbf{y}$—i.e., to infer which concepts are active in an activation vector. When $d_z > d_y$, more concepts are encoded than there are activation dimensions, a regime known as *superposition* (Elhage et al., 2022). The system $\mathbf{y} = \mathbf{W}\mathbf{z}$ is underdetermined: each observation $\mathbf{y}$ is consistent with infinitely many $\mathbf{z}$, so one cannot simply learn a linear unmixing to map activations back to concepts. In other words, linear *representation*—concepts being linearly encoded in activations—is not the same as linear *accessibility*—that concepts are recoverable by a linear transformation. However this distinction is routinely overlooked and the LRH is conflated with the much stronger latter claim.

Concept recovery requires additional structure to resolve the underdetermination, such as by assuming sparsity, i.e., in practice, only $k \ll d_z$ concepts are active in any given input. Compressed sensing is precisely the framework that characterises when sparse signals can be recovered from such underdetermined measurements (Donoho and Elad, 2003; Donoho, 2006a). Under the sparsity assumption, classical results show that k-sparse codes can be recovered from $d_y = \mathcal{O}(k \log(d_z/k))$ input dimensions via nonlinear algorithms (e.g., basis pursuit, iterative thresholding), while requiring recovery to be *linear* (e.g., a sparse linear probe) increases the required number of dimensions to $d_y = \Omega_\epsilon\left(\frac{k^2}{\log k} \log\left(\frac{d_z}{k}\right)\right)$ (Garg et al., 2026), a quadratic blowup in k. Concretely, for $d_z = 10^6$ latent concepts with $k = 100$ active at once, nonlinear recovery requires only $d_y \approx 920$ dimensions while linear recovery requires $d_y \approx 20{,}000$—a typical transformer

hidden size of 4096 comfortably exceeds the former but falls far short of the latter. Thus, whether concepts are recoverable from a fixed set of activations therefore depends not on whether they are linearly encoded, but on *how* one attempts to decode them.

Additionally, with respect to downstream implications, the compression through $\mathbf{W}$ folds distinct regions of the latent space onto one another, so that a decision boundary that is linear in $\mathbf{z}$-space can become nonlinear in $\mathbf{y}$-space (Figure 1). Recovering $\mathbf{z}$ from $\mathbf{y}$ therefore requires nonlinear methods, but which nonlinear method matters. *Amortised* inference (Gregor and LeCun, 2010), as implemented by Sparse Autoencoders (SAEs) (Ng et al., 2011; Cunningham et al., 2023), learns a fixed encoder $r : \mathbb{R}^{d_y} \to \mathbb{R}^{d_h}$ that maps an activation to a sparse code in a single forward pass by training on a finite distribution of training data. *Per-sample inference*, by contrast, solves an optimisation problem from scratch for each input, using only the observation $\mathbf{y}$ and the dictionary $\mathbf{W}$, with no dependence on a training distribution.

Amortisation gap. The solution learned via amortised or per-sample inference can be evaluated by how accurately either recovers the true sparse code, e.g., whether the correct concepts are identified as active. In-distribution, both approaches may recover $\mathbf{z}$ well, and the discrepancy between them—the *amortisation gap* (Margossian and Blei, 2023; Kim and Pavlovic, 2021; Zhang et al., 2022; Cremer et al., 2018; Schott et al., 2021; Paiton et al., 2020; O'Neill et al., 2024)—can be small. Under distribution shift, however, the gap widens: when the sparsity pattern changes (e.g., novel combinations of concepts co-activate), the amortised encoder's recovery degrades because it was optimised for training-time statistics, while per-sample methods that optimise from scratch for each input remain unaffected. This paper studies the amortisation gap in the context of interpretability under superposition:

> *Under what conditions does sparse inference recover the true latent factors from superposed activations, and how does the choice of inference procedure—amortised, per-sample, or hybrid—affect robustness under distribution shift?*

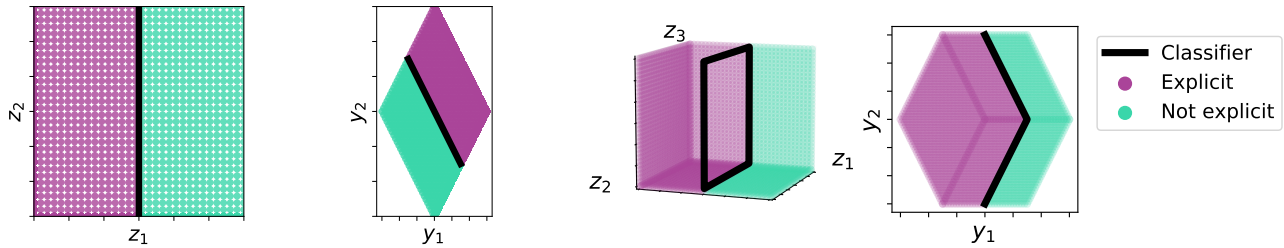
This argument adds nuance to recent studies suggesting that linear probes trained on top of LLM activations are superior to SAEs in out-of-distribution (OOD) binary classification tasks (Kantamneni et al., 2025). We argue that the recent OOD failures of SAEs are not an indictment of the superposition hypothesis, but rather a predictable consequence of replacing principled sparse inference with a brittle, amortised encoder. Instead of discarding the powerful framework of sparse coding, we embrace the geometric consequences of superposition and utilize methods equipped to handle the nonlinearity it induces.

Main contributions. In this work, we revisit classical sparse coding to address the central question. We show that under superposition, even labels that are linearly separable in latent space may become nonlinearly separable in activation space. We demonstrate that this is particularly pronounced in OOD settings, so that a perfect linear probe trained in-distribution will fail OOD (Figure 1). SAEs perform nonlinear inference, but amortising it into a fixed encoder introduces a systematic amortisation gap (Figure 5): the encoder fits the training distribution's co-occurrence structure and fails to generalise to novel combinations of latent factors under OOD composition shift (Figure 3). An oracle baseline—per-sample FISTA with the ground-truth dictionary—achieves near-perfect OOD recovery at all scales tested, proving the problem is solvable under compressed sensing theory (Section 4). Through controlled experiments that decompose the SAE failure, we identify dictionary learning—not the inference procedure—as the binding constraint. SAE-learned dictionaries point in substantially wrong directions, and replacing the encoder with per-sample FISTA on the same dictionary does not close the gap (Figure 8). Classical dictionary learning (DL-FISTA) produces better dictionaries at small scale, but both methods fail at dictionary learning when the latent dimension grows large. These results reframe the SAE failure: the bottleneck is not amortisation of inference but amortisation of dictionary learning, and the path forward requires scalable algorithms for learning dictionaries under the compressed-sensing framework.

2 RELATED WORK

Compositional generalisation—the ability to understand and produce novel combinations of learned concepts—remains a fundamental challenge for neural networks (Fodor and Pylyshyn, 1988; Hupkes et al., 2020). Current approaches in causal representation learning attempt to achieve this through structural constraints, such as *additive* decoders (Lachapelle et al., 2023) (e.g., in SAEs), or specific training objectives like compositional risk minimisation (Mahajan et al., 2025). These works focus on obtaining guarantees for the compositional generalisation of disentangled models, while here, we evaluate the effect of compositional shifts under superposition.

SAEs have recently emerged as a primary tool for decomposing the internal activations of LLMs into interpretable, monosemantic features (Cunningham et al., 2023). Despite their success in interpretability, their out-of-distribution (OOD) robustness is a growing concern. Recent evaluations suggest that SAEs trained on general datasets often fail to discover generalisable concepts across different domains or layers (Heindrich et al., 2025), underperform compared to simple linear probes (Kantamneni et al., 2025), and remain brittle even when scaled (Gao et al., 2024). Interestingly, this brittleness is less pronounced in domain-specific applications, such as medical QA or pathology, where SAEs have shown more



(a) (b)

Figure 1: Binary classification with $t = \mathbf{1}_{z_1} > 0.5$ (green: not explicit, purple: explicit). **(a)** When $d_z = d_y$, the linear decision boundary in latent space remains linear after mixing $\mathbf{y} = \mathbf{W}\mathbf{z}$. **(b)** When $d_z > d_y$ (overcompleteness) and z sparse, we can project down into non-overlapping regions (i.e., compressed sensing is possible), but the decision boundary becomes nonlinear in activation space, making linear probes insufficient.

stable and biologically relevant feature transfer (O'Neill et al., 2025; Le et al., 2024). These conflicting results motivate a more principled evaluation of SAEs under distribution shifts (Joshi et al., 2025).

Recent work has highlighted the limitations of SAEs in recovering true latent variables (O'Neill et al., 2024; Paulo and Belrose, 2025). This stands in contrast to the classical sparse coding framework (Olshausen and Field, 1996; Ranzato et al., 2007), which utilises iterative optimisation rather than a learned encoder to recover latent variables. This iterative approach provides stronger theoretical guarantees for the unique recovery of latents (Hillar and Sommer, 2015; Lewicki and Sejnowski, 1997; Gribonval et al., 2015). In contrast to O'Neill et al. (2024), who explore different in-distribution (ID) amortisation strategies, we focus our analysis on downstream tasks under OOD compositional shifts.

Lastly, literature on overcomplete independent component analysis (Podosinnikova et al., 2019; Wang and Seigal, 2024) explores identifiability where the number of latent variables exceeds the number of observed variables ($d_z > d_y$), though these models typically rely on statistical independence rather than sparsity.

3 AMORTISATION VS POINTWISE SPARSE INFERENCE AND COMPOSITIONAL GENERALISATION

We begin by specifying the data-generating process. Consider latent variable vectors $\mathbf{z} \in \mathbb{R}^n$ with at most k non-zeros. A support set $S \subseteq [n]$ is drawn first to index these non-zero components following the process:

$$\begin{aligned} S \sim p_S, \quad z \sim p(z \mid S) \\ \operatorname{supp}(p_S) \subseteq \mathcal{S}_k := \{S \subseteq [n] : |S| \leq k\}, \\ \mathbb{P}(\mathbf{z}_{S^c} = \mathbf{0} \mid S) = 1. \end{aligned} \tag{1}$$

An unknown generative process g maps latents to data, producing $\mathbf{x} := g(\mathbf{z}) \in \mathbb{R}^{d_x}$. Although $\mathbf{z}$ is unobserved, we have access to learned representations $\mathbf{y} := f(\mathbf{x}) \in \mathbb{R}^{d_y}$ for some fixed (encoding) function $f : \mathbb{R}^{d_x} \to \mathbb{R}^{d_y}$ (e.g., an LLM activation map). Since the coordinates of $\mathbf{y}$ need not necessarily align with those of $\mathbf{z}$, we fit a representation model $r : \mathbb{R}^{d_y} \to \mathbb{R}^{d_h}$ estimating $\mathbf{h} := r(\mathbf{y})$ and another decoder $q : \mathbb{R}^{d_h} \to \mathbb{R}^{d_y}$ s.t. $\hat{\mathbf{y}} := q(\mathbf{h})$. Ideally, $\mathbf{h}$ serves as a proxy for $\mathbf{z}$ and when d_z is known, we can set $d_h = d_z$, while $d_y < d_z$. We refer to individual coordinates of $\mathbf{h}$ (one-dimensional subspaces of $\mathbb{R}^{d_h}$) as *features*, and we seek features that correspond (approximately) to the latent coordinates $\mathbf{z}$.

Typically, $\mathbf{h}$ is enforced to be sparse, in line with Equation (1). In practice, SAEs (Cunningham et al., 2023) are commonly used to learn the autoencoder $q \circ r$, where typically the decoder q is assumed to be linear and the encoder r is a single-linear layer followed by an activation function such as ReLU (Cunningham et al., 2023), JumpReLU (Rajamanoharan et al., 2024), or TopK (Gao et al., 2024; Costa et al., 2025). The assumption of a linear decoder is motivated by the LRH stating that the composed map $f \circ g$ is linear in the underlying latent variables s.t.,

$$\mathbf{y} \;=\; f(g(\mathbf{z})) \;\approx\; \mathbf{A}\mathbf{z} + \mathbf{b}, \qquad \mathbf{x} \sim p_{\mathbf{x}}, \tag{2}$$

for some matrix $\mathbf{A} \in \mathbb{R}^{d_y \times d_z}$ and offset $\mathbf{b} \in \mathbb{R}^{d_y}$, where $p_{\mathbf{x}}$ denotes the induced data distribution under the generative process $\mathbf{x} = g(\mathbf{z})$. A long line of work provides evidence for this hypothesis (c.f. Rumelhart and Abrahamson (1973); Hinton et al. (1986); Mikolov et al. (2013); Ravfogel et al. (2020); Klindt et al. (2025)). More recently, theoretical work justifies why linear properties could arise in these models (c.f. Jiang et al. (2024); Roeder et al. (2021); Marconato et al. (2024); Reizinger et al. (2024)).

Ensuring injectivity under overcompleteness. SAEs are often trained with an overcomplete feature dimension $d_h > d_y$, so that the decoder operates as a linear dictionary $\mathbf{W} \in \mathbb{R}^{d_h \times d_z}$. When $d_h > d_y$, the decoder dictionary $\mathbf{W} \in \mathbb{R}^{d_y \times d_h}$ has more columns than rows, i.e. it is projecting from a high to a low-dimensional space, like projecting a 3D object onto its 2D shadow. This compression means there must